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Evaluation of the Safety and Pharmacokinetics of Single-Dose Oral Probenecid Administration in Healthy Dogs

Margaret Cook | Daniel L. Gustafson | Katie Kroeker | Sarah Shropshire | Kristin M. Zersen

Department of Clinical Sciences, College of Veterinary Medicine and Biomedical Sciences, Colorado State University, Fort Collins, Colorado, USA

Correspondence: Kristin M. Zersen (kristin.zersen@colostate.edu)

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Keywords: bioavailability | grape toxicity | OAT-1 inhibitor | raisin toxicity | tartaric acid

ABSTRACT

Background: Grape-induced acute kidney injury (AKI) is caused by tartaric acid and may lead to death in dogs. Probenecid, an organic anion transporter-1 inhibitor, recently has been shown to block the uptake of tartaric acid in Madin–Darby canine kidney cells and has been suggested as a possible target for prevention of AKI after grape ingestion.

Hypothesis/Aims: Assess the safety and pharmacokinetics (PK) of PO probenecid in dogs. We hypothesized that probenecid would result in mild, self-limiting gastrointestinal (GI) adverse effects and would be safe in healthy dogs. Additionally, we hypothesized that PO probenecid (50 mg/kg) would have adequate bioavailability and achieve pharmacologically active plasma drug concentrations.

Animals: Six healthy beagle dogs.

Methods: Pharmacokinetic (PK) study. Dogs were given 50 mg/kg of probenecid PO, with PK data collected for 48 h after administration. Complete blood count, serum biochemistry profile, urinalysis, and clinical monitoring were performed throughout a 21-day study period to assess safety. Plasma concentration versus time data was analyzed using non-compartmental and two-compartmental modeling.

Results: Orally administered probenecid had excellent estimated bioavailability (82.6%) and rapid absorption, with a mean maximal plasma concentration of 589.3 μM (range: 368.0–830.5 μM) within 1.5 h. The mean volume of distribution was 0.71 L/kg, with mean systemic clearance of 0.022 L/h/kg and mean half-life of 24.1 h. Probenecid was well tolerated by all six dogs, with no clinically relevant adverse effects noted.

Conclusions and Clinical Importance: Orally administered probenecid is safe and bioavailable in healthy dogs. Future clinical trials assessing PO probenecid in dogs with known tartaric acid ingestion are warranted.

Abbreviations: AKI, acute kidney injury; AST, aspartate aminotransferase; AUC, area under the curve; AUMC, area under the moment curve; CK, creatine kinase; C_{Max}, peak plasma concentration; GI, gastrointestinal; LLOQ, lower limit of quantification; NAC, non-compartmental; nRBC, nucleated red blood cell; OAT-1, organic anion transporter-1; OAT-4, organic anion transporter-4; PK, pharmacokinetic; T_{max}, time to maximum plasma concentration.

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1 | Introduction

The first report of grape-induced acute kidney injury (AKI) in dogs was made over 20 years ago [1]. Aside from decontamination by induction of emesis and hospitalization for IV fluids, no specific treatment or antidote is available for grape toxicity [2]. The prevalence of AKI after ingestion of grapes is difficult to estimate because most dogs do not present to the veterinarian after ingestion. However, in one study of 120 dogs that presented to veterinarians with known grape ingestion, 6.7% (8/120) developed AKI, and one dog (12.5%, 1/8) with AKI died [3]. In 2022, tartaric acid was identified as the likely toxic component in grapes [4]. Tartaric acid is transported from the blood into canine proximal tubular cells via organic anion transporter-1 (OAT-1). It is hypothesized that in humans, the organic anion transporter-4 (OAT-4) secretes tartaric acid into the tubular fluid by which it is excreted from the body in the urine. However, dogs lack OAT-4 in their kidneys, which prevents the transport of tartaric acid into the urine, resulting in accumulation in the proximal tubular cells and leading to species-specific renal toxicity after grape ingestion in some dogs [5].

Probenecid, an OAT-1 inhibitor, recently has been shown to block the uptake of tartaric acid in Madin–Darby canine kidney cells, preventing in vitro cytotoxicity [5]. This discovery has prompted interest in probenecid as a potential therapeutic target for preventing grape toxicity in dogs. If probenecid blocks the uptake of the toxin into proximal tubular cells, it may be given to dogs immediately after grape ingestion and preclude hospitalization and additional treatment.

Although the pharmacokinetics (PK) of IV probenecid in dogs has been reported, the PK and safety of PO probenecid have not been evaluated in dogs [6]. Our aim was to assess the PK and safety of PO probenecid in healthy dogs. We hypothesized that PO probenecid (50 mg/kg) would have adequate bioavailability and achieve pharmacologically active plasma drug concentrations. Our second hypothesis was that PO probenecid would result in only mild, self-limiting adverse gastrointestinal (GI) effects and would be deemed safe in healthy dogs.

2 | Materials and Methods

2.1 | Animals

Six healthy research beagle dogs were used in the study. Food was withheld for 12 h before probenecid administration, and water was always available. Dogs resumed their normal feeding schedule on the evening of Day 0. On Day 0, each dog was given approximately 50 mg/kg probenecid (Actavis, Parsippany-Troy Hills, New Jersey) PO with approximately 1 tablespoon of canned dog food. Dogs were monitored at least once daily throughout the 21-day study period for clinical signs including vomiting, diarrhea, changes in appetite, abdominal pain, joint pain, changes in urinary habits, and lethargy. Vital signs (temperature, heart rate, respiratory rate) were measured at least once daily from Day 0 to 4, and body weight was recorded on Days 0, 7, 14, and 21.

2.2 | Sample Collection and Processing

After PO probenecid administration, 1.5 mL of venous blood was collected by venipuncture and placed into a tube without additives for PK analysis at the following time points: 0 (immediately before probenecid administration), 0.5, 1, 2, 4, 8, 16, 24, and 48 h post-probenecid administration. Within 1 h of blood collection, samples were centrifuged at 3200 g for 10 min, and serum was separated and stored in a −80°C freezer until batch analysis was performed.

To assess the safety of PO probenecid, venous blood (3.0 mL) and urine (1.0–5.0 mL obtained by cystocentesis or voiding) were collected for CBC, serum biochemistry profile, and urinalysis on Days 0, 1, 3, 7, 14, and 21. Blood samples for CBC, biochemistry panel, and urinalysis were submitted for analysis within 1 h of collection.

2.3 | Analytical Assay

Plasma probenecid concentrations were determined using liquid chromatography coupled to tandem mass spectrometry (LCMS). The LCMS-based methods used for probenecid analysis in plasma have been reported previously [6]; these methods were adapted for use in our study [6].

Samples were analyzed using an ABI 6500 Q-TRAP triple quadrupole mass spectrometer (Applied Biosystems Inc., Foster City, CA) coupled to a Shimadzu LC-30 AD SIL HPLC liquid chromatography system with an integrated autosampler (Shimadzu Scientific Instruments Inc., Colombia, MD) used for analysis. Probenecid (MedChem Express—HYB0545R) and internal standard naringenin (NGN; Cayman Chemical—14173) were obtained as lyophilized powder and stored according to manufacturer recommendations. Stock solutions of probenecid and NGN were made at 1 mg/mL in acetonitrile. Assay performance based on quality control (QC) sample analysis at 20, 200, and 2000 ng/mL showed an accuracy and precision (CV%) of 91.2% ± 6.8%. Batch acceptance was based on 83% (15/18) QC samples having an accuracy > 85%, and the lower limit of quantification (LLOQ) for the analysis was 4.6 ng/mL based on the lowest concentration at least 2-fold above baseline and having an accuracy > 75%.

An 11-point calibration curve was prepared in canine plasma using probenecid at 10-fold concentrations (50, 25, 12.5, 6.25, 3.125, 1.56, 0.78, 0.39, 0.195, 0.098, and 0.046 μg/mL). One hundred microliters of plasma was spiked with 10 μL of 10× standard PBN plus 10 μL of 10× internal standard (NGN at 2 μg/mL), followed by the addition of 400 μL of cold acetonitrile. Eighteen samples were prepared as six replicates each for low (20 ng/mL), medium (200 ng/mL) and high (2000 ng/mL) QC samples in the same manner. For samples with concentrations outside of the upper limit of quantification (50 000 ng/mL), samples were diluted in canine plasma before analysis.

2.4 | PK Analysis and Statistical Analysis

Plasma probenecid concentration versus time data was evaluated using nonlinear regression in a two-compartment model,

using Phoenix WinNonLin v 8.3.4.295 (Certara, Princeton, NJ). A non-compartmental (NAC) analysis also was applied to model the PK data. The area under the curve (AUC) and area under the moment curve (AUMC) were calculated using a linear trapezoidal method with linear interpolation. Normality of data was assessed using a Shapiro–Wilk test.

Both approaches to modeling the PK data were performed to provide descriptive data (without assumptions related to drug disposition) and to allow for predictions of probenecid disposition characteristics.

3 | Results

Six healthy intact male beagle dogs were used in the study. Median age was 7.7 years (range: 7.3–9.2 years) and median body weight was 9.4 kg (range: 9.0–10.4 kg). All were assessed as healthy based on the results of the initial physical examination, CBC, serum biochemistry profile, and urinalysis.

On Day 0, each dog was given approximately 50 mg/kg (median: 53.2 mg/kg; range: 50.0–55.5 mg/kg) probenecid by mouth with approximately 1 tablespoon of canned dog food.

3.1 | PK of Orally Administered Probenecid

Probenecid was first detected at the 0.5-h time point in all dogs. Figure 1 presents a graphic presentation of non-compartmental model of plasma probenecid concentration over time for all six dogs (Figure 1A) and a two-compartment model for all six dogs at each time point (Figure 1B). Peak plasma concentration (C_{max}) was either at the 1- or 2-h time point for all six dogs (Table 1). Using data from a previous study [6], the PO bioavailability of probenecid in dogs was estimated to be 82.6%.

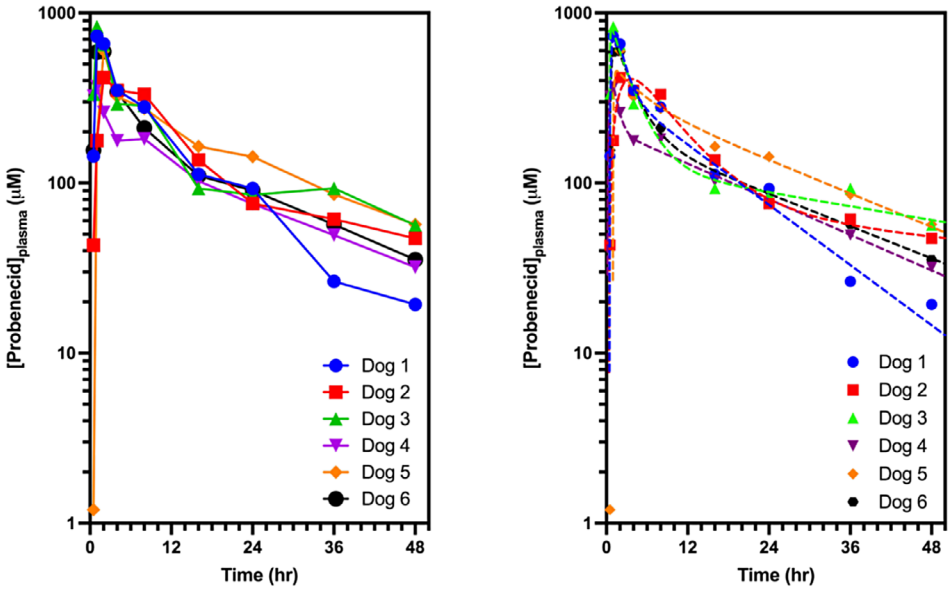

FIGURE 1 | Graphic presentation of the non-compartmental model of probenecid plasma concentrations over time (h) for six dogs, each dog represented by a different color. (A) Graphic presentation of the two-compartment model for six dogs over time (h), each dog represented by a different color (B).

TABLE 1 | Non-compartmental and two-compartment model of pharmacokinetics of oral probenecid (50 mg/kg) in six healthy dogs.

Pharmacokinetic parameters of orally dosed probenecid (50 mg/kg) by non-compartmental and two-compartment analysis in six healthy dogs			
Parameter	**Units**	**Non-compartmental**	**Two compartment**
C_{Max}	µM	589.3 ± 176.9 (368.0–830.5)	575.5 ± 190.7 (373.3–819.4)
T_{max}	h	1.5 ± 0.5 (1–2)	1.6 ± 1.1 (0.8–3.7)
AUC_{0-48h}	µM h	6611 ± 1027 (4810–7767)	—
AUC_{0-inf}	µM h	8215 ± 2095 (5895–11 636)	8397 ± 2477 (5453–11 758)
V_z/F	L/kg	0.71 ± 0.25 (0.41–1.13)	—
CL_s/F	L/h/kg	0.022 ± 0.005 (0.015–0.031)	0.023 ± 0.007 (0.015–0.032)
Half-life	h	24.1 ± 14.2 (11.4–51.9)	31.1 ± 25.5 (10.2–77.3)

Abbreviations: AUC_{0-48h}, area under the plasma concentration–time curve from time zero to 48 h; AUC_{0-inf}, area under the plasma concentration–time curve from time zero to infinity; CL_s/F, apparent clearance; C_{max}, peak plasma concentration; T_{max}, time to maximum plasma concentration; V_z/F, apparent volume of distribution (adjusted for bioavailability).

3.2 | Safety Assessment of Orally Administered Probenecid

3.2.1 | Complete Blood Count

Hematologic evaluation included standard CBC variables. Throughout the 21-day study period, no clinically relevant abnormalities were observed in the myeloid cell lines or in the platelet counts for all six dogs. Despite the absence of anemia, the nucleated red blood cell (nRBC) count was inconsistently increased in five of six dogs. All increases were considered mild, with the highest nRBC count being $0.2 \times 10^3/\mu L$ (reference interval: $0–0.001 \times 10^3/\mu L$) [7].

3.2.2 | Biochemistry

All six dogs experienced one or more increases in creatine kinase (CK) activity, with or without a concurrent increase in aspartate aminotransferase (AST). The highest CK activity recorded among the dogs was 1054 IU/L (reference range: 70–275 IU/L). The highest AST activity recorded was 60 IU/L (reference range: 15–45 IU/L). No other clinically relevant biochemistry abnormalities were noted throughout the 21-day study period. A summary of laboratory test values outside of the reference range is presented in Table 2.

3.2.3 | Urinalysis

Urine samples could not be collected at 5 time points, because the bladder was too small for collection by cystocentesis and the dog would not voluntarily urinate. The urine specific gravity of all six dogs remained > 1.019 (range: 1.019–1.065) throughout the 21-day study period. Sporadic bacteriuria was noted in 3 dogs, but several of the urine samples (2/4 samples) had been collected by voiding. The dogs with sporadic bacteriuria did not have any clinical signs of lower urinary tract disease at those time points. No other clinically relevant urinalysis abnormalities were observed.

3.2.4 | Monitoring and Observation

After PO administration of probenecid, no clinically relevant abnormalities in vital signs were recorded for any of the six dogs. No vomiting, changes in appetite, abdominal pain, joint pain, lethargy, or changes in urination were observed throughout the 21-day study period. No clinically relevant changes in body weight occurred during the 21-day study period.

All six of the dogs developed soft feces (Nestlé Purina fecal score: 3–6/7) during Days 2 and 3 of the study. The soft feces resolved without medical treatment, with a Nestlé Purina fecal score ≤ 3/7 by Day 4 of the study in all dogs [8]. None of the dogs became dehydrated, and none appeared clinically affected by the soft feces.

4 | Discussion

Administration of 50 mg/kg probenecid PO was found to be safe in healthy dogs, with only mild, self-limiting soft feces being noted, supporting our hypothesis. Additionally, the estimated bioavailability was high (82.6%). Probenecid has been evaluated previously for its ability to decrease renal clearance of ranitidine in dogs [9]. In that study, the C_{max} of probenecid to achieve this pharmacologically active effect on ranitidine renal clearance was approximately 140 μM [9]. We found a mean C_{max} of 589.3 μM, well above the plasma concentration noted previously [9], thus demonstrating that pharmacologically active concentrations were reached. In a previous in vitro study in which Madin–Darby kidney cells were exposed to 500 and 1000 μM concentrations of probenecid, the uptake of tartaric acid into the kidney cells was blocked [5]. The mean maximum plasma concentration of probenecid in our study was 589.3 μM, which suggests that the uptake of tartaric acid could be effectively blocked. However, there are important differences between this study and the previously referenced study. Specifically, the previously referenced study was a cellular incubation study in which the cells were exposed to a constant concentration of probenecid for 72h [5]. Because the previous study was an in vitro model and because of differences in timing, direct comparisons between the studies cannot be made. Future studies are indicated to determine the plasma probenecid concentration required to block the uptake of tartaric acid and determine how long such concentrations must be maintained.

Probenecid is commonly prescribed to humans for treatment of gout, and it may be prescribed with antibiotics to increase blood concentrations of antibiotics [10]. In humans, PO probenecid achieves plasma concentrations identical to those achieved after IV administration, indicating 100% bioavailability [10]. Peak plasma probenecid concentration after PO administration in humans occurs between 1 and 9 h, and half-life is dose dependent, ranging from 4 to 12 h [10]. With IV administration in humans, probenecid has a volume of distribution of 11 L with a dose-dependent half-life of 2–12 h [10]. It is eliminated by renal excretion, independent of dose administered, and excretion depends on pH and rate of urine flow [10].

The PK data of probenecid has been described for several veterinary species. The PK of IV probenecid have been previously reported in dogs, with a half-life of 17.7 h, a steady-state volume of distribution of 0.46 L/kg, and a total body clearance of 0.34 mL/min/kg (or 20.4 mL/h/kg) reported [6]. When administered PO to horses, the half-life of probenecid was 2 h, with a time to maximal plasma concentration (T_{max}) of 45 min, and bioavailability of 91.20% [11].

To supply descriptive data, without assumptions related to probenecid disposition, the data was fit to a NAC PK model. The mean half-life was 24.1 h, with plasma concentrations still detectable 48 h after drug administration. Probenecid was rapidly absorbed after PO administration, with a C_{Max} of 589.3 μM and T_{max} average of 1.5 h. Using the PO PK data from our study and IV PK data from a previous study [6], the PO bioavailability of probenecid in dogs was estimated to be 82.6%.

In summary, PO bioavailability is excellent in humans (100%), the horse (91.2%), and the dog (82.6%) [10, 11]. In our study, PO probenecid had a rapid T_{max} in dogs (1.5 h) when compared to the dose-dependent range in humans (1–9 h) [10, 11]. More

TABLE 2 | Number of healthy dogs receiving oral probenecid with bloodwork values outside the reference interval at each sampling time point.

	Reference interval	Number of dogs with value outside reference interval [actual values]					
		Day 0	Day 1	Day 3	Day 7	Day 14	Day 21
HCT	40%–55%	0	1 [57]	0	0	1 [56]	0
RETIC	$0–100 \times 10^3/\mu L$	0	0	0	2 [108.7, 100.5]	2 [102.5, 108.1]	0
nRBC	$0–0.001 \times 10^3/\mu L$	0	2 [0.2,0.1]	3 [0.1, 0.2, 0.1]	3 [0.1, 0.1, 0.1]	0	0
NCC	$4.5–15.0 \times 10^3/\mu L$	0	0	1 [16.9]	0	0	0
NEUT	$2.6–11 \times 10^3/\mu L$	0	0	1 [15]	0	0	0
LYMPH	$1–4.8 \times 10^3/\mu L$	0	0	0	0	0	2 [0.7, 0.9]
EOS	$0.1–1.2 \times 10^3/\mu L$	0	0	0	0	0	0
PLAT	$200–500 \times 10^3/\mu L$	0	1 [517]	0	0	0	0
GLU	70–115 mg/dL	1 [118]	0	0	1 [65]	0	3 [117, 127, 123]
BUN	7–30 mg/dL	0	0	0	0	0	0
CREAT	0.6–1.6 mg/dL	1 [0.41]	2 [0.51, 0.46]	1 [0.51]	2 [0.5, 0.43]	2 [0.39, 0.56]	1 [0.47]
PHOS	2.5–6.0 mg/dL	0	0	0	0	0	0
Ca^{2+}	9.0–11.5 mg/dL	0	0	0	0	0	0
TP	5.0–7.0 g/dL	0	0	0	0	0	0
ALB	3.0–4.3 g/dL	0	0	0	0	0	0
GLOB	1.5–3.2 g/dL	0	0	0	2 [3.9, 3.8]	0	0
T. BILI	0.0–0.2 mg/dL	0	0	0	0	0	0
ALP	15–140 IU/L	0	0	0	0	0	0
ALT	10–90 IU/L	0	0	1 [92]	0	0	0
AST	15–45 IU/L	0	2 [60, 47]	1 [55]	0	3 [52, 58, 48]	0
CK	50–275 IU/L	0	3 [363, 457, 533]	2 [284, 516]	1 [310]	4 [801, 1054, 605, 440]	3 [358, 324, 381]
Na^+	142–152 mEq/L	0	0	0	0	0	0
K^+	3.9–5.4 mEq/L	0	0	0	0	0	0
Cl^-	108–118 mEq/L	0	0	0	1 [107.9]	0	1 [106.7]

Note: Abnormal values recorded for each dog are denoted within brackets.
Abbreviations: ALB, albumin; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; CA^{2+}, total calcium; CK, creatine kinase; Cl^-, chloride; CREAT, creatinine; EOS, eosinophils; GLOB, globulins; GLU, blood glucose; HCT, hematocrit; K^+, potassium; LYMPH, lymphocytes; NA^+, sodium; NCC, nucleated cell count; NEUT, neutrophils; nRBC, nucleated red blood cell; PHOS, phosphorus; Retic, reticulocytes; T. BILI, total bilirubin; TP, total protein.

research is warranted to determine if T_{max} in dogs is dose-dependent or fixed. Of the three species, the horse has the most rapid half-life (2 h) with PO administration of probenecid [11]. Humans have a dose-dependent half-life range (4–12 h) and dogs have the longest half-life (24.1 h) [10]. The half-life in dogs in our study was similar to the half-life after IV administration in dogs (17.7 h) [6].

In addition to ranitidine, probenecid previously has been given to dogs to determine its effect on the renal clearance of bumetanide, indocyanine green, and phenolsulfonphthalein [9, 12–15]. In those studies, dogs were given 7.5–100 mg/kg probenecid IV and no adverse effects were reported, but the studies did not explicitly state that adverse effects and complications were not seen [9, 12–15]. Probenecid is considered to have minimal risk for toxicity, with minimal to no hemopoietic, renal, or hepatic toxicity reported in humans [16]. The most common adverse effects of probenecid in humans include nausea, vomiting, abdominal cramping, fever, rash, myalgia, and headaches [16].

For our study, we selected a PO dose of 50 mg/kg, which fell in the middle of the previously described IV dose ranges [9, 12–15]. No complications or adverse effects occurred in our study, supporting our second hypothesis that PO probenecid is safe in healthy dogs. Transient soft feces were recorded as the only clinical effect in all six dogs. None of the dogs had clinical evidence

of dehydration, and no medical interventions were required before the soft feces resolved spontaneously.

The changes in laboratory test results observed throughout the 21-day study required no medical intervention. The nRBC, a precursor of the reticulocyte, is normally found in bone marrow and usually only seen in peripheral blood in the presence of anemia [17]. In the absence of anemia, nRBCs in low numbers can be seen in the peripheral blood of healthy dogs [17]. The mild increases in nRBC counts seen in five of the six dogs in our study were considered to be clinically unimportant and possibly not related to the administration of probenecid.

The CK activities were normal in all six dogs before administration of probenecid but increased at different time points in all six dogs. The time of increase in CK activity was inconsistent across all six dogs. The half-life of CK is only 2.5 h, and increased activity may not have been associated with probenecid administration, but rather a consequence of multiple blood collections required for the study, handling of the dogs during the study, or a consequence of musculoskeletal impact during social play in housed groups [18]. The increases in CK activity detected are considered to be clinically unimportant and unlikely to be related to the administration of probenecid. However, myalgia is a reported adverse effect in humans, and thus mild muscle injury cannot be ruled out.

Probenecid currently is prescribed only as a PO medication for humans [16]. Because the PO formulation is the only commercially available option, we selected PO probenecid for our study design. Furthermore, in dogs with grape exposure, PO administration of probenecid may be preferred over IV administration, because the goal would be to use it as an outpatient treatment. In our study, soft feces were noted in all dogs, and thus concurrent probiotic or fiber-based medications or supplements could be considered in clinical patients, but the potential effect of these interventions on probenecid is unknown.

A limitation of our study is the small number of dogs treated with a single dose of the drug. Small sample size decreases the ability to assess variability among individuals, apply the PK data to different breeds and sexes, or apply the data to a population of dogs with systemic illness (e.g., renal injury). Furthermore, a control group was not included, limiting our ability to assess if changes in laboratory test results were associated with probenecid administration or not.

Our results raise new questions on how probenecid can be used in dogs in a clinical setting. If effective in clinical patients, treatment with probenecid may minimize morbidity and mortality in dogs exposed to tartaric acid. Future clinical studies evaluating PO administration of probenecid in dogs with naturally occurring tartaric acid exposure should be considered.

5 | Conclusion

We demonstrated that 50 mg/kg probenecid PO is bioavailable and safe in healthy dogs. A recent in vitro study determined that probenecid can successfully block cellular uptake of tartaric acid, the toxin associated with grapes or raisins, by Madin–Darby renal cells of dogs [5]. Our results show that PO probenecid potentially could be administered to block in vivo uptake of nephrotoxic tartaric acid, preventing renal injury in dogs. Future clinical trials of administration of probenecid to dogs with known tartaric acid ingestion are warranted.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by the Institutional Animal Care and Use Committee (IACUC) of Colorado State University (IACUC #5896). Authors declare human ethics approval was not needed. Animals were cared for according to the principles outlined in the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. S. Gwaltney-Brant, J. K. Holding, C. W. Donaldson, P. A. Eubig, and S. A. Khan, "Renal Failure Associated With Ingestion of Grapes or Raisins in Dogs," *Journal of the American Veterinary Medical Association* 218, no. 10 (2001): 1555–1556.

2. J. E. Hoffber, "Grapes and Raisins," in *Blackwell's Five-Minute Veterinary Consult Small Animal Toxicology*, 2nd ed., ed. L. R. Hovda, A. G. Brutlag, R. H. Poppenga, et al. (Wiley Blackwell, 2016), 485–490.

3. C. F. Reich, M. C. Salcedo, A. M. Koenigshof, et al., "Retrospective Evaluation of the Clinical Course and Outcome Following Grape or Raisin Ingestion in Dogs (2005–2014): 139 Cases," *Journal of Veterinary Emergency and Critical Care* 30, no. 1 (2020): 60–65.

4. C. A. Wegenast, I. D. Meadows, R. E. Anderson, T. Southard, C. R. González Barrientos, and T. A. Wismer, "Acute Kidney Injury in Dogs Following Ingestion of Cream of Tartar and Tamarinds and the Connection to Tartaric Acid as the Proposed Toxic Principle in Grapes and Raisins," *Journal of Veterinary Emergency and Critical Care* 32, no. 6 (2022): 812–816.

5. S. R. Coyne and G. M. Landry, "Tartaric Acid Induces Toxicity in Madin–Darby Canine Kidney Cells, but Not Human Kidney-2 Cells In Vitro, and Is Prevented by Organic Anion Transporter Inhibition and Human OAT-4 Transfection," *Journal of Veterinary Emergency and Critical Care* 33, no. 3 (2023): 298–304.

6. T. Kakizaki, Y. Yokoyama, M. Natsuhori, et al., "Probenecid: Its Chromatographic Determination, Plasma Protein Binding, and In Vivo Pharmacokinetics in Dogs," *Journal of Veterinary Medical Science* 68, no. 4 (2006): 361–365.

7. Colorado State University Clinical Pathology Laboratory, "Reference Intervals Established Using Quality Assurance and Laboratory

Standards Guidelines Per the American Society for Veterinary Clinical Pathology," (2024).

8. C. L. Cavett, M. Tonero, S. L. Marks, J. A. Winston, C. Gilor, and A. J. Rudinsky, "Consistency of Faecal Scoring Using Two Canine Faecal Scoring Systems," *Journal of Small Animal Practice* 62, no. 3 (2021): 167–173.

9. S. P. Boom, I. Meyer, A. C. Wouterse, and F. G. Russel, "A Physiologically Based Kidney Model for the Renal Clearance of Ranitidine and the Interaction With Cimetidine and Probenecid in the Dog," *Biopharmaceutics & Drug Disposition* 19, no. 3 (1998): 199–208.

10. R. F. Cunningham, Z. H. Israili, and P. G. Dayton, "Clinical Pharmacokinetics of Probenecid," *Clinical Pharmacokinetics* 6 (1981): 135–151.

11. J. M. Donecker, R. A. Sams, and S. M. Ashcraft, "Pharmacokinetics of Probenecid and the Effect of Oral Probenecid Administration on the Pharmacokinetics of Cefazolin in Mares," *American Journal of Veterinary Research* 41, no. 1 (1986): 89–95.

12. H. S. H. Lau, L.-J. Shih, and D. E. Smith, "Effect of Probenecid on the Dose-Response Relationship of Bumetanide at Steady State," *Journal of Pharmacology and Experimental Therapeutics* 227, no. 1 (1983): 51–54.

13. F. G. Russel, A. C. Wouterse, and C. A. van Ginnekaen, "Physiologically Based Pharmacokinetic Model for Renal Clearance of Phenolsulfonphthalein and the Interaction With Probenecid and Salicyluric Acid in the Dog," *Journal of Pharmacokinetics and Biopharmaceutics* 15, no. 4 (1987): 349–368.

14. E. E. Vogin, W. Scott, J. Boyd, W. T. Bear, and P. A. Mattis, "Effect of Probenecid on Indocyanine Green Clearance," *Journal of Pharmacology and Experimental Therapeutics* 152, no. 3 (1966): 509–515.

15. S. D. Holland and H. E. Williamson, "Probenecid Inhibition of Bumetanide-Induced Natriuresis in the Dog (40540)," *Proceedings of the Society for Experimental Biology and Medicine* 161 (1979): 299–302.

16. N. Robbins, S. E. Koch, M. Tranter, and J. Rubinstein, "The History and Future of Probenecid," *Cardiovascular Toxicology* 12, no. 1 (2012): 1–9.

17. J. W. Harvey, "Evaluation of Erythrocytes," in *Veterinary Hematology: A Diagnostic Guide and Color Atlas*, 1st ed., ed. J. W. Harvey (Elsevier, 2012), 49–121.

18. M. Aktas, H. P. Lefebvre, P. L. Toutain, and J. P. Braun, "Disposition of Creatine Kinase Activity in Dog Plasma Following Intravenous and Intramuscular Injection of Skeletal Muscle Homogenates," *Journal of Veterinary Pharmacology and Therapeutics* 18, no. 1 (1995): 1–6.